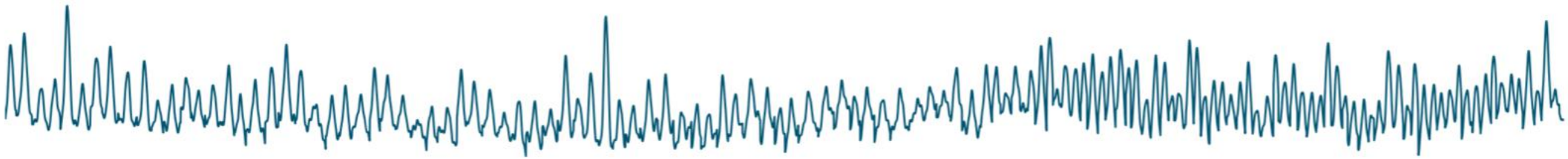
Noéa

Reading water as rhythm, memory and mood

Affective computing for water systems



Hong Kong, 1884–2025: one beat per year. Height is the year's peak daily rainfall; spacing is the time since the previous peak.

MEETING OF WATERS

Data can show that a river is changing. It rarely makes anyone feel it.

Noéa works in the missing layer: turning measurement into meaning people can experience.

It reads a record the way a cardiologist reads an ECG

01

A long record

Daily rainfall or river discharge, covering decades to centuries.

02

Diagnostics

Peaks and the gaps between them, return periods, flow duration, recession, the seasonal cycle.

03

Two affective values

Valence: pleasant to distressing.

Arousal: calm to activated.

Alongside four physical readouts: size, glow, texture, tempo.

04

XR experience

Designed to feed an immersive XR experience. The output format is still provisional.

Built to be universal

Each record fits its own model on its own history. Another rainfall or river record can go through the same pipeline.

Hong Kong rainfall	Public record, Hong Kong Observatory. Daily, 1884–2025.
Rhine and alpine rivers	Public records, GRDC and FOEN. Daily, 8 stations.
The next record	Not yet run. Tested on two records so far.

Two layers, kept deliberately apart

Diagnostic layer

PLAIN SCIENCE

Hong Kong rainfall, 1884–2025: annual peaks, the gaps between them, return periods.

Rhine and alpine rivers, 8 stations: flow duration, recession, the seasonal cycle.

Measures what happened, and is never adjusted for artistic reasons.

Curatorial layer

DESIGNED

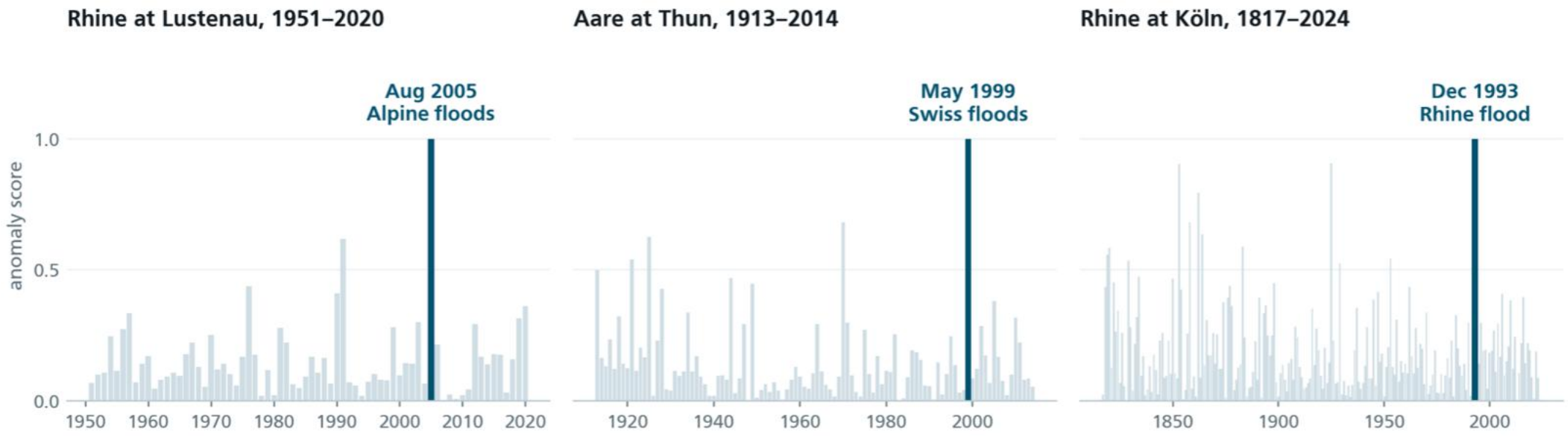
Decides how the results should feel: how alarming, how heavy.

Currently kept separate from the measurements, so the feeling could be redesigned without changing any of the science.

Its output: valence and arousal.

Machine learning finds unusual years

Its one job so far, unsupervised: learn what a typical year looks like, then score how badly each year fits.



The highlighted bar is each station's highest-scoring year. The model was never told these events existed.

Method: PCA reconstruction error, fit separately per station on its own years. Events: Aug 2005 Alpine floods (22–23 Aug), May 1999 Swiss floods (Thun, 15 May), Christmas 1993 Rhine flood (Köln, 24 Dec).

The end result is how people respond

Noéa connects machine intelligence to natural intelligence: a model learns the rhythm of a river or rainfall record and flags when it breaks.

How people respond to that now has to be tested in human labs, with HCI methods. That is the research.